

CO3- AT3- Tree Traversal Challenges

Write the traversal sequences for following trees

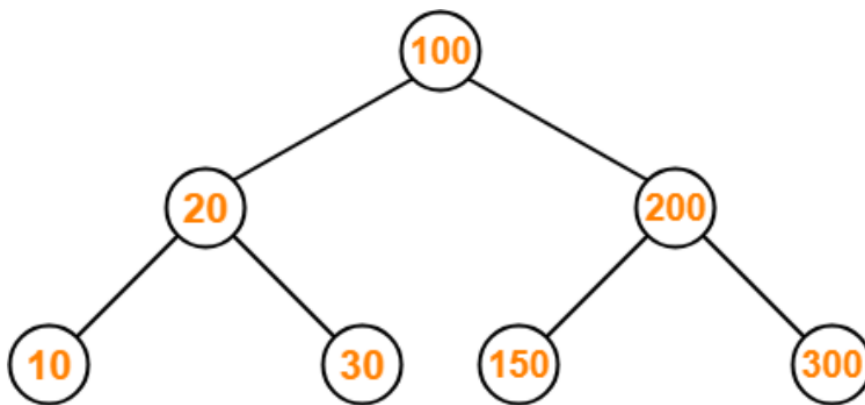
CSA 0315 – Data Structures

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Question No :1

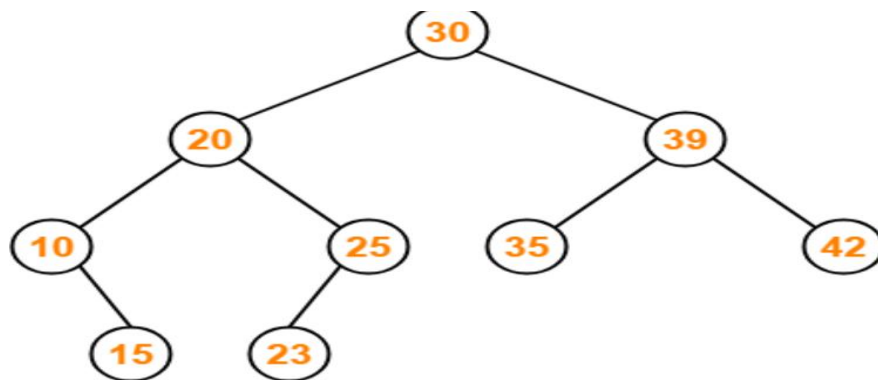


Inorder: 10,20,30,100,150,200,300

Preorder: 100,20,10,30,200,150,300

Postorder: 10,30,20,150,300,200,100

Question No : 2



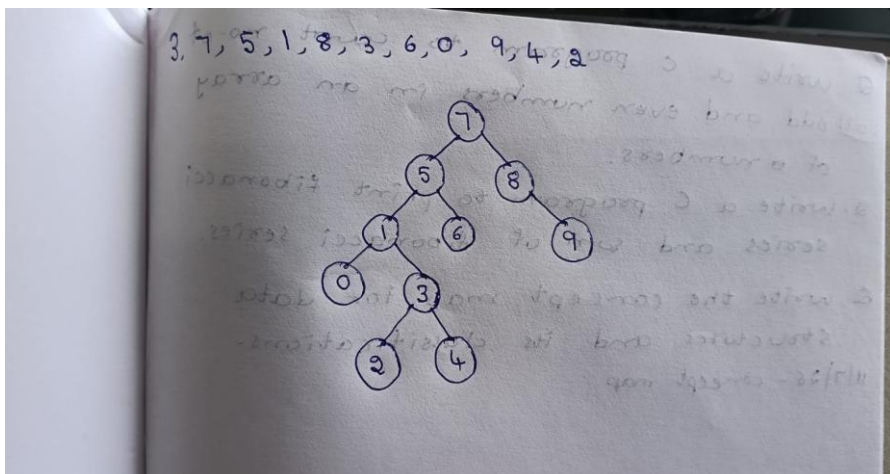
Inorder: 10,15,20,23,25,30,35,39,42

Preorder: 30,20,10,15,25,23,39,35,42

Postorder: 15,10,23,25,20,35,42,39,30

Question No: 3

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?



Inorder: 0,1,2,3,4,5,6,7,8,9

Preorder: 7,5,1,0,3,2,4,6,8,9

Postorder: 0,2,4,3,1,6,5,8,9,7

Question No: 5

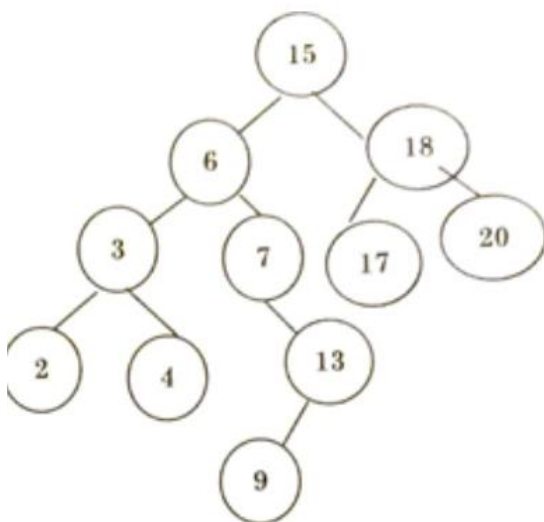


Inorder: 4,2,7,5,1,3,6

Preorder: 1,2,4,5,7,3,6

Postorder: 4,7,5,2,6,3,1

Question No 6



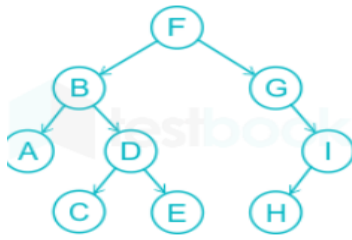
Inorder: 2,3,4,6,7,13,9,15,17,18,20

Preorder: 15,6,3,2,4,7,13,9,18,17,20

Postorder: 2,4,3,7,13,9,6,17,20,18,20

Question No 7

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G

2. A, C, D, E, B, F, G, H, I

3. A, B, C, D, E, F, G, H, I

4. I, H, G, F, E, D, C, B, A

3.A,B,C,D,E,F,G,H,I

Question No 8

Consider the following Inorder and Preorder traversal of a tree

Inorder – D B E F A G H C

Preorder - A B D E F C G H

What is the Postorder Traversal of the same tree

1. D F E B H C G A

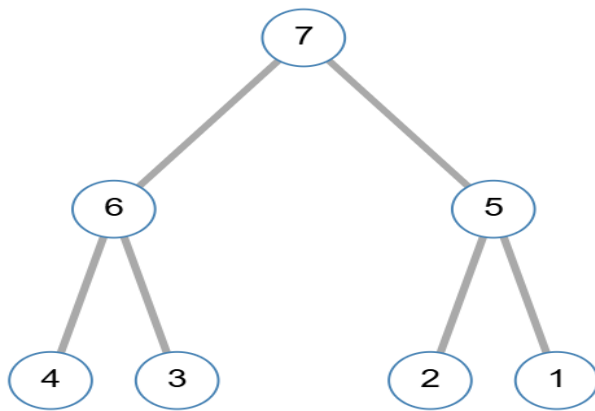
2. D F E B G H C A

3. D F E B H G C A

4. D F E H B G C A

3.D,F,E,B,H,G,C,A

Question No 9: Find Inorder, Preorder and Postorder

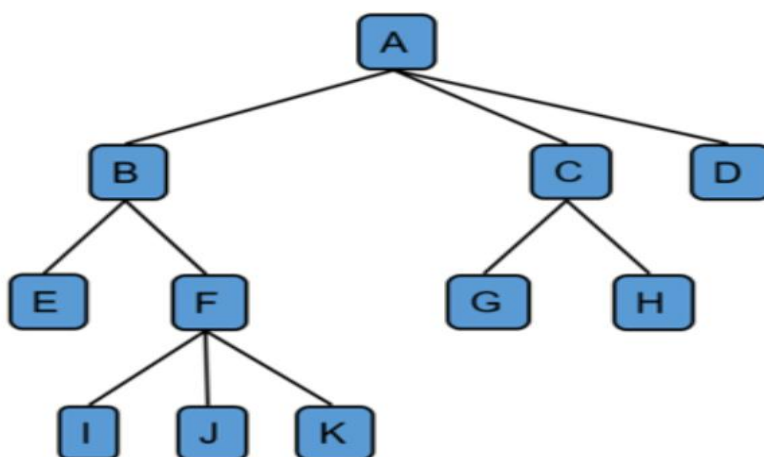


Inorder: 4,6,3,7,2,5,1

Preorder: 7,6,4,3,5,2,1

Postorder: 4,3,6,2,1,5,7

Question No 10



Inorder: E,B,I,J,F,A,G,H,C

Preorder: A,B,E,F,I,J,C,G,H

Postorder: E,I,J,FB,G,H,C,A